Setup Guide

From zero to the app running on your phone, then recording the demo.

Your time	About 60–90 minutes, one time
Cost	$0 for Supabase and Expo. About $5 of Claude API credit covers the whole demo with room to spare.
What's already built	The full app, database, security rules, AI functions and tests. You are only connecting accounts.
Deadline	**12:00 pm EDT (noon), Monday Oct 26, 2026**, per the official 2026 rules. Confirm that your district is participating.

Already online: `https://ashih-10.github.io/Pepper/` runs the app as an on-device demo: no account needed, and everything stays in the visitor's browser. Join the sample group with code `CHEM24`. The steps below set up the real app, with real accounts and live group sync between phones.

Two rules that protect your account: (1) The Claude API key only ever goes into Supabase secrets (Step 6). Never put it in the app, in `.env`, or in GitHub. (2) Never paste keys into a chat with any AI, including me.

What you need first

Account / tool	Why	Where
Node.js 20+ and Git on your laptop	Runs the app's dev server	nodejs.org, git-scm.com
Supabase account	Database, sign-in, live group sync, AI functions	supabase.com (free tier)
Claude Console account	The API key for screenshot reading + game plan	console.anthropic.com
Expo Go app on your phone (and a friend's)	Runs the app without the App Store	App Store / Google Play
Google Cloud project (optional)	Only for the "Continue with Google" button	console.cloud.google.com

1 Get the code and install

```
git clone https://github.com/ashih-10/<your-repo>.git
cd <your-repo>
npm install
```

2 Create the Supabase project

1. supabase.com → **New project**. Pick a region near you and save the database password somewhere safe.
2. When it finishes, open **Project Settings → API Keys**. Copy the **Project URL** and the **publishable key** (starts with `sb_publishable_`).
3. In the code folder: copy `.env.example` to `.env` and paste both values in. The publishable key is safe in the app; the security rules protect the data.

3 Create the database tables and security rules

```
npx supabase login
npx supabase link --project-ref <your-project-ref>   # the ref is in your project URL
npx supabase db push
```

This runs `supabase/migrations/…_init.sql`: every table, the group rules, and row-level security. **Fallback** if the CLI gives you trouble: open **SQL Editor** in the dashboard, paste that whole file, and press Run.

4 Sign-in settings

1. **Authentication → Sign In / Providers → Email**: turn **off** "Confirm email" for the demo, so new accounts work immediately. **Turn it back on after the competition, but connect a real email provider first:** while it's off, anyone can make unlimited accounts, which weakens the join-code and AI-spending limits (see the Security Review). Supabase's built-in email only reaches members of your Supabase team (2 an hour), so set up Resend (free) under **Authentication → SMTP Settings** before turning confirmation on, or real students won't get their emails.
2. **Authentication → URL Configuration → Redirect URLs**: add `exp://**` and `http://localhost:8081`. Only needed for Google sign-in.

5 Claude API key with a spending limit

1. Claude Console → **Billing**: add $5 of credit.
2. **Limits**: set a monthly spend limit (for example $10). This is your safety net if a bug ever loops.
3. **API Keys → Create key**. Copy it for the next step only.

What it costs: one assignment photo on Claude Haiku is roughly a tenth of a cent; one game plan on Claude Sonnet is roughly one cent. The app also caps each student at 40 AI calls a day.

6 Store the key on the server and deploy the AI functions

```
npx supabase secrets set ANTHROPIC_API_KEY=<paste key here>
npx supabase functions deploy extract-task
npx supabase functions deploy plan-today
```

Type the key straight into your terminal. Then clear it from your clipboard.

7 Run it on your phone

```
npx expo start
```

- **iPhone:** scan the QR code with the Camera app. **Android:** scan it inside Expo Go.
- Phone and laptop must be on the same Wi-Fi. School Wi-Fi often blocks this; use `npx expo start --tunnel` instead.
- Press `w` in the terminal to open the web version. That's an easy second account for the demo (groups, AI and sync all work on web; pepper *alerts* only work on phones).

8 Optional: "Continue with Google"

Email sign-in is enough for the demo. If you want Google: create an OAuth client (type **Web application**) in Google Cloud, add Supabase's callback URL (shown on **Authentication → Sign In / Providers → Google**) as an authorized redirect URI, then paste the client ID and secret into that Supabase page. Budget an hour; skip it if you are short on time.

9 Put the web version online with Vercel

> **Fastest way to see it working (about 20 minutes):** do Steps 2, 3 and 4, then this step. Steps 5 and 6 (the AI key) can come after; until then the photo and game-plan buttons show an error and everything else works.

1. vercel.com → sign up with GitHub → **Add New → Project** → import **ashih-10/Pepper**.
2. Leave the build settings alone: the repo's `vercel.json` already sets them.
3. Open **Environment Variables** and add the same two values as your `.env`: `EXPO_PUBLIC_SUPABASE_URL` and `EXPO_PUBLIC_SUPABASE_PUBLISHABLE_KEY`.
4. Press **Deploy**. After about 2 minutes you get a link like `pepper-xxxx.vercel.app`.
5. In Supabase, **Authentication → URL Configuration**: set **Site URL** to that link and add it to Redirect URLs.

The two values are baked into the build. If you change them later, press **Redeploy** in Vercel. Every push to GitHub redeploys automatically. Pepper alerts only work in the phone app; groups, live sync and AI all work on the web. If the app says "Demo mode" instead of showing a sign-in form, the environment variables are missing or misspelled.

Why Vercel and not the GitHub Pages link for the real app: every GitHub Pages site on your account shares one web address, so any of your other Pages sites could read the app's sign-in session. Vercel gives the real app its own address. The Pages link is fine for the demo, where nothing private is stored.

Recording the demo

This follows the demo script in your PRD. Before you start: on **Me**, tap **Load a demo week** and turn on **Demo clock**.

#	Show	How
1	The problem in one line	Say it to camera: deadlines are everywhere and nothing tells you what matters today.
2	Onboarding	Make a fresh account: add classes, pick colors, answer the goals question.
3	Photo → task	Tap the camera button, photograph a real assignment sheet, show the ⚠️ fields and the AI's question, confirm at the bottom.
4	Peppers heating up	On Today, tap **+12h** / **+1 day** on the demo clock and watch homework climb from 🫑 to 👻.
5	Groups	Second account joins with the code. The same shared test shows 🔥 Habanero for the busy student and 🌶️ Serrano for the other. Move the date and show the alert and the activity feed.
6	Milk	Check off a task. Milk pours, the pepper cools, the task is struck through.

> **Day before recording and day before judging:** open your Supabase dashboard. Free projects pause after about a week without activity. Also tap **Remove demo tasks** on Me when you're done.

If something goes wrong

Symptom	Fix
App says "Demo mode" when you expected sign-in	`.env` (or the Vercel environment variables) is missing or has a typo. Fix it, then restart with `npx expo start -c` (or Redeploy).
Expo Go can't connect	Use `npx expo start --tunnel`.
Sign-up says "check your email"	Turn off "Confirm email" (Step 4).
"The AI had a problem"	Check the key was set (Step 6) and the Console has credit. Logs: Supabase → Edge Functions → the function → Logs.
"You've hit today's AI limit"	40 calls per student per day. It resets at midnight UTC.
No alerts on the phone	Allow notifications for Expo Go in phone settings. On Me, tap **Send a test alert**.
Friend's iPhone can't open it	Make sure their Expo Go is up to date, or use the web version for the second account.

Running the tests

```
npm test                # 58 tests: pepper formula, recurring tasks, alerts, demo backend
npx supabase start      # needs Docker Desktop
npm run test:db         # 27 security attack tests on a real local database
```